

REQUIREMENTS ENGINEERING

LECTURE 6: REQUIREMENTS SPECIFICATION

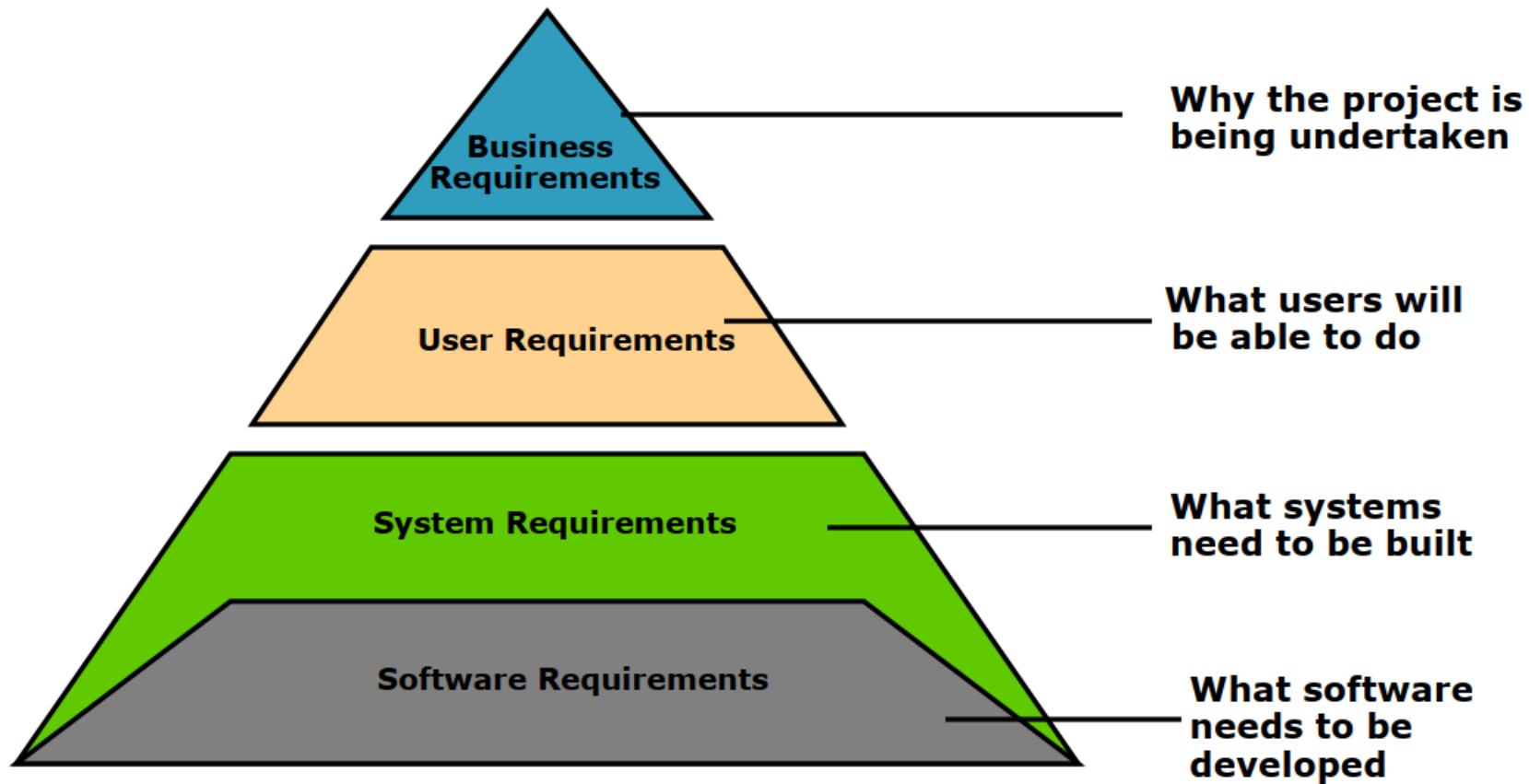


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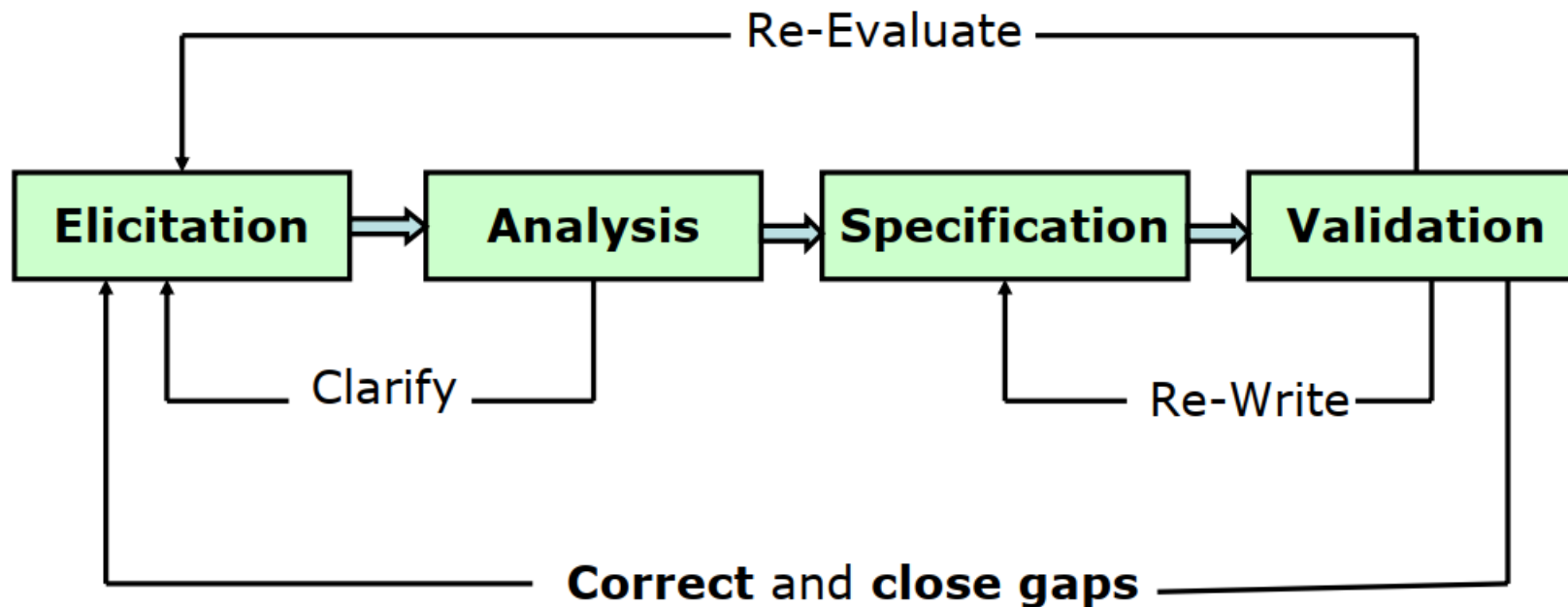


Năm học: 2021 -2022

Types of requirement



Requirements Development Process



Iterative process - Multiple-steps, Not sequential



Document Requirements

- Requirements engineering is a **communication intensive** activity.
- Communication should **start as early** as possible since stakeholders may have **different ideas** of exactly what “requirements” are.
- The most essential practice is to **write down** the requirements in a structured format as you gather and analyze them.



Requirements Specification

- The techniques of **documenting** the external behavior of a system that will be built based on the features selected during the analysis process.
- Requirements specification is the process of **elaborating, refining, and organizing requirements into documentation.**
- The development of a document that **clearly and precisely records** each of the requirements of the system or product.



Audiences

- Decision Boards (Business Manager, Project Manager,....)
- Users
- Developers, Programmers
- Architect
- Testers



Good requirements

- **Clear** Avoid confusion
- **Brief** Short and simple
- **Verifiable** Testable and verifiable
- **Traceable** Uniquely identified, can be tracked



Software Requirement Specification (SRS)

- The software requirements specification (SRS) is a **precise record of requirements** that enables software developers to design, develop, and test the software solution.
- Its purpose is to document **functional requirements, quality attributes, constraints and external interfaces** for the software solution. It also serves **as a contract** between customers and software organizations.



Requirements Specification in Agile

- In agile methodologies such as extreme programming (XP) or Scrum formal, static documentation such as a software requirements specification (SRS) are usually replaced by a more lightweight documentation of the requirements, namely **Product Backlog**.
- The **agile product backlog** is in the form of **user stories**, which are short, simple descriptions of the desired functionality told from perspective of the user.
- When applying Scrum, it's **not necessary to start a project with a lengthy**, upfront effort to document **all requirements**.
- The Scrum product backlog is **allowed to grow and change** as more is learned about the product and its customers.

Thank you
It's time for your questions

